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25 August 1954
SUPERSEDING
Army 91-130
28 February 1948

MILITARY SPECIFICATION

CLAMP, HOSE AND AIR DUCT (TANK AND AUTOMOTIVE)

1. SCOPE

1.1 This specification covers hose and air duct clamps for use on vehicle and other military engines.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-P-416	- Plating, Cadmium (Electrodeposited)
QQ-Z-325	- Zinc Plating (Electrodeposited)
ZZ-H-428	- Hose; Coolant-System, for Motor Vehicles

Military

MIL-B-833	- Belts, Waist, Web, with Clip, without Buckle
MIL-V-3329	- Vehicles, Combat, Self-Propelled and Towed; General Specification
MIL-H-6000	- Hose, Aircraft (Fuel, Oil, Coolant, Water and Alcohol)
MIL-P-11443	- Packaging and Packing for Shipment and Storage of Spare Parts for Military Vehicles
MIL-L-2105	- Lubricant, Gear, Universal

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

Army Ordnance Corps

CMAX-10	- Clamp, Hose and Duct (Commercial)
CMAX-10.1	- Clamp, Hose and Duct (Commercial)

DRAWINGS

Any drawings supplied or approved by the procuring agency.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency, or as directed by the contracting officer.

3. REQUIREMENTS

3.1 General.-Except as modified herein, or otherwise specified, clamps shall conform to applicable provisions of Specification MIL-V-3329.

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3.2 Material, treatments, and finishes.

3.2.1 Materials. - Materials used in the construction of clamps covered by this specification shall be as specified herein and on applicable standards or drawings. Steel or alloy steel shall conform to requirements specified herein, on standards or drawings, or in procurement documents, as to classification, finish, edge, and other physical and chemical properties and shall assure conformance to requirements of this specification and reliable performance in intended use. Materials not specifically designated shall be suited to intended use.

3.2.2 Protective finishes. - Parts made of metals or alloys that are not inherently corrosion resistant shall be electroplated in conformance to Class A of Specification QQ-P-416 (or, if specificed, to class I of Specification QQ-Z-325). Special treatments of inherently corrosion-resistant alloys, if specified or required, shall be such as to assure conformance to requirements herein.

3.3 Design and construction. - Design and construction shall conform to applicable standards and drawings, with dimensions and clamping ranges as specified, and shall assure satisfaction of all requirements herein.

3.4 Ease of installation. - Each clamp, unless otherwise specified, shall remain a single assembly, with no separable or loose parts when opened for installation around a hose, and when tightened after such installation shall continue to satisfy all requirements of this specification. Installing, tightening, loosening, and removing shall require no special tools.

3.5 Free torque. - Wrench torque required to turn the adjuster (worm, screw, bolt, or nut) back and forth through the entire range of adjustment, when measured in conformance to 4.5, shall at no point exceed 3 pound-inches.

3.6 Torque capacity. - When tested as specified in 4.6, clamps shall withstand five applications of 50 pound-inches torque without failure of any part of the clamp and thereafter shall satisfy 3.7 and 3.9.

3.7 Release torque. - When measured as specified in 4.7, the torque required to release the adjuster immediately after it has been tightened to 30 pound-inches shall be not less than 8 pound-inches.

3.8 Strap ductility. - When subjected to 180-degree bend around a 3/16-inch diameter mandrel as specified in 4.8, and straightened, the band shall show no break, crack, or other indication of failure.

3.9 Leakage pressure. - When measured as specified in 4.9, no leakage shall occur at specified torques at test pressures specified in table I.

Table I. - Leakage pressures

Hose inside diamater, nominal (inches)	Leakage pressure, p.s.i. at applied torque of	
	20 lb-in.	40 lb-in.
5/16 and less	600	600
3/8 to 1/2	470	600
1/2	350	600
5/8	280	350
3/4 to 1	235	350
1	175	350
1 1/8 to 1 1/2	150	300
1 3/4 to 2	125	200
2 1/4 to 4	100	200
4 1/4 to 6	75	175

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3.10 Durability. - After clamp adjuster has been tightened to 40 pound-inches torque and immediately released 15 times under test conditions specified in 4.10, the parts of the clamp shall show no deformation or excessive wear and shall meet the requirements of 3.7 and 3.11.

3.11 Damage to hose or duct. - When clamps are tested as specified in 4.10 and 4.11, the repeated tightening and loosening of the clamp shall not seriously damage the outer covering of the hose, the materal of the duct, or the webbing used in the test. Cutting through outer covering of the hose or cutting halfway thorugh the exposed yarns of duck or webbing shall be regarded as serious damage.

3.12 Prelubrication. - Prior to or during clamp assembly, the worm, or adjuster threads, shall be given at least a light coating of lubricant conforming, unless otherwise specified, to grade 90 of Specification MIL-L-2105.

3.13 Marking for identification. - Each clamp shall be marked as specified on applicable standards or drawings, or on procurement documents, to show part number, and name or approved trade mark of manufacturer.

3.14 Workmanship. - Workmanship shall assure a product conforming to all requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.

4.1.1 Lot size. - Unless otherwise specified, a lot shall consist of all hose or duct clamps of one part number, from one manufacturer, submitted at one time for acceptance.

4.1.2 Sampling for visual, dimensional, and primary functional inspection. - The Government inspector shall select for inspection (see 4.2.2) a representative sample from each lot in conformance to inspection level II of Standard MIL-STD-105.

4.1.3 Sampling for test. - From the sample specified in 4.1.2 the Government inspector shall select, at random, not fewer than four representative clamps for each test specified for acceptance in 4.3.1.

4.1.3.1 Specimen identification. - Prior to any tests, each specimen shall be suitably marked for identification in a manner that will in no way damage the clamp or impair its performance.

4.2 Inspection.

4.2.1 Place of inspection. - Unless otherwise specified, inspection shall be conducted at the place of manufacture.

4.2.2 Visual, dimensional, and primary functional inspection. - Hose and duct clamps selected as specified in 4.1.2 shall be inspected for conformance to the following acceptable quality levels:

Classification of defects	Acceptable quality level percent defective
Major	1.0
Minor	2.5

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4.2.2.1 Classification of defects.-For inspection, defects shall be classified as follows:

Major defects (AQL-1.0)
Dimensions (see 3.3)

Minor defects (AQL-2.5 percent)
Free torque (see 3.5)
Lubrication (see 3.12)
Marking (see 3.13)
Workmanship (see 3.14)

4.3 Tests.-

4.3.1 Acceptance.-Specimens selected in accordance with 4.1.3 shall be subjected to the tests specified in 4.6 to 4.11 inclusive and to such other tests as the Government inspector deems necessary to determine conformance to this specification. When test equipment of the correct size is not available for the leakage test (see 4.9) and if specifically approved by the procuring activity, clamps of a different size but of the same construction and materials may be tested in lieu of the clamps presented for acceptance.

4.3.2 Rejection.-Failure of a single specimen in any test shall be cause for rejection of the lot or, at the option of the Government inspector, for retest of twice the original number of specimens. Failure of more than one specimen in the original test or failure of any retest specimen shall result in rejection of the lot.

4.4 Test conditions and equipment.-Tests, except as otherwise specified, shall be performed with equipment and ambient air at a temperature between 60° and 80°F., torque and pressure measuring and indicating devices shall be accurate to within ± 5 percent at values indicated. Test mandrels (or heavy-wall tubes) shall conform in general to figure 1, unless equivalent equipment is approved (see 6.3).

4.5 Free torque.-To ascertain conformance to 3.5, the torque indication of a suitable torque wrench shall be watched while the worm or adjuster is turned slowly back and forth between the specified limits of clamp adjustment.

4.6 Torque capacity.-To ascertain conformance to 3.6, clamp shall be installed over hose conforming to Specification ZZ-H-428, or to Specification MIL-H-6000, over a mandrel (or heavy-wall tube or approved disc, see 4.4 and 6.2) of proper size for the hose, both hose and mandrel sizes to be proper for the size of the clamp to be tested (see 4.9). The worm or adjuster shall be tightened to specified torque with a suitable torque wrench and immediately loosened, specified number of times, and shall thereafter be carefully inspected for any damage to clamp parts. Clamps having a minimum clamping capacity greater than 4 inches may be tested, unless otherwise specified, by clamping webbing, 1 1/4 inch, 2 ounces per lineal yard conforming to Specification MIL-B-833 (see 6.1 and 6.2), wrapped around suitable solid mandrel, thick disc, or pipe or tube of diameter within specified clamping rande of clamp tested (see 3.3 and 6.2). Thereafter clamps shall be subjected to tests specified in 4.9.

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4.7 Release torque. - To ascertain conformance to 3.7, five new clamps shall be installed over hose (or webbing) and mandrel as specified in 4.6 and each clamp tightened to specified torque and immediately released five times, with release torque recorded for each release and values averaged,

4.8 Strap ductility. - To ascertain conformance to 3.8, the clamp band shall be carefully strightened, then bent, cold, 180 degrees, in the direction of the original bend, around a 3/16 inch diameter mandrel, then released and restraightened. Careful examination shall be made while bent, after spring-back, and after restraightening, for evidence of damage. Two bends shall be made in each perforated strap, one in the perforated portion and the other as close as practicable midway between the nearest perforation and the nearest edge of the saddle.

4.9 Leakage pressure. - To ascertain conformance to 3.9 and table I, clamps of 4-inch and less minimum clamping capacity (see 3.3) shall be installed over 3-inch lengths of clean, dry hose conforming to Specification ZZ-H-428 or Specification MIL-H-6000 fitted, without lubicant or sealant, over a clean, smooth test mandrel, or tube, conforming in general to figure 1 (see 6.3). Clamps shall be tightened with torque-indicating wrench to 20 pound-inch, and later to 40 pound-inch torque, and the test pressures specified in table I for each torque applied while careful watch is kept for leakage.

4.10 Endurance. - To ascertain conformance to 3.10, at least four clamps of the size tested shall be installed over hose or webbing in manner specified in 4.6, and tightened to specified torque and loosened the specified number of times, after which careful inspection shall be made for deformation or wear. Thereafter tests 4.7 and 4.9 shall be made.

4.11 Damage to hose or duct. - To ascertain conformance to 3.11, hose, duct, or webbing, clamped as specified in 4.10, shall be examined carefully after the fifth, tenth, and fifteenth loosening of the clamp (during test 4.10), for evidence of damage to the surface or of cutting or tearing of the clamped material.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging.

5.1.1 For immediate use at first domestic receiving activity. - Unless otherwise specified clamps prepared for level A packing shall be cleaned, preserved, and packaged in conformance to best commercial practice.

5.1.2 For entry into the supply system. - Except as otherwise specified, clamps for level B or C packing shall be cleaned and packaged in conformance to Specification MIL-P-11443, except that no preservative is requred, and that, unless otherwise specified, a unit package may contain any number of clamps up to 50, all of same size and part number.

5.2 Packing.

5.2.1 Level A-domestic shipment-immediate use. - When level A packing is specified (see 6.2) clamps shall be packed in conformance to best commercial practice.

5.2.2 Level B-domesitc shipment-limited storage. - When level B packing is specified (see 6.2) shall be packed as specified for domestic shipment in Specification MIL-P-11443.

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5.2.3 Level C-overseas shipment or extended storage.-When level C is specified (see 6.2), clamps shall be packed as specified for oversea shipment in Specification MIL-P-11443.

5.3 Marking.

5.3.1 Marking of interior packages and exterior containers.-Interior packages and exterior container shall be marked as specified in the applicable requirements of Standard MIL-STD-129 except, the following marking shall be added as applicable, and in conformance to any special instructions in the procurement documents:

5.3.1.1 For level A packing.-Each package or container shall be clearly marked "PACKAGED FOR IMMEDIATE USE" or "PACKED FOR IMMEDIATE USE" as applicable.

5.3.1.2 For level B packing.-Each package or container shall be clearly marked "PACKAGED FOR LIMITED STORAGE" or "PACKED FOR LIMITED STORAGE" as applicable.

6. NOTES

6.1 Intended use.-Clamps covered by this specification, of sizes up to and including 5 1/8- or 6 1/8-inch maximum clamping capacity (outside diameter of hose), will be used largely for clamping hose of cooling systems of military vehicle and other military engines (power plants), hose for heated-liquid-type vehicle heating equipment, and some semiflexible connectors in induction manifolds and elsewhere in engine induction systems. Larger sizes, and some of the smaller, when required, are used for clamping or securing flexible ducts (usually of fabric, treated or impregnated fabric, or of rubber or rubber-like materials) used for conducting heated or cool air. Clamps specified herein are not intended for use in installations requiring strength exceeding that of clamps conforming, to this specification and to the dimensions and materials specified on the applicable standards and drawings.

6.2 Ordering data.-Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Ordnance part number and applicable drawings. (See 2.1.)
c. Packing level required. (See 5.2.)
d. Any special marking requirements. (See 5.3.)

6.3 Test equipment.-Satisfactory test mandrels, (see Figure 1 and paragraphs 4.4, 4.6, and 4.9) can be made from pipe or tubing having wall thickness sufficient to resist specified clamping pressures without excessive distortion, with suitable tubing and needed fittings for conducting fluid pressure to the circumferential groove. Surface finish of new mandrels where in contacts with clamped test hose should be between 32 and 63 microinches r.m.s. After removal of test hose, surface smoothness can be restored effectively by polishing carefully with suitable abrasive cloth of proper grain size used with oil (120 or 150 for final polishing, 80 or 100 for rough finishing). Hard-surfaced or chromium electrodeposited mandrels have been found to require less attention to maintain desired surface smoothness. Clamping tests can be made satisfactorily by clamping hose (or webbing) around circular discs 1 inch or more thick of metal or other rigid material. Such discs, of metal, can be used as "plugs" in ends of 3-inch lengths of hose (see 4.9) for leakage tests, one such cylindrical plug to have suitable fitting for admission of fluid under pressure.

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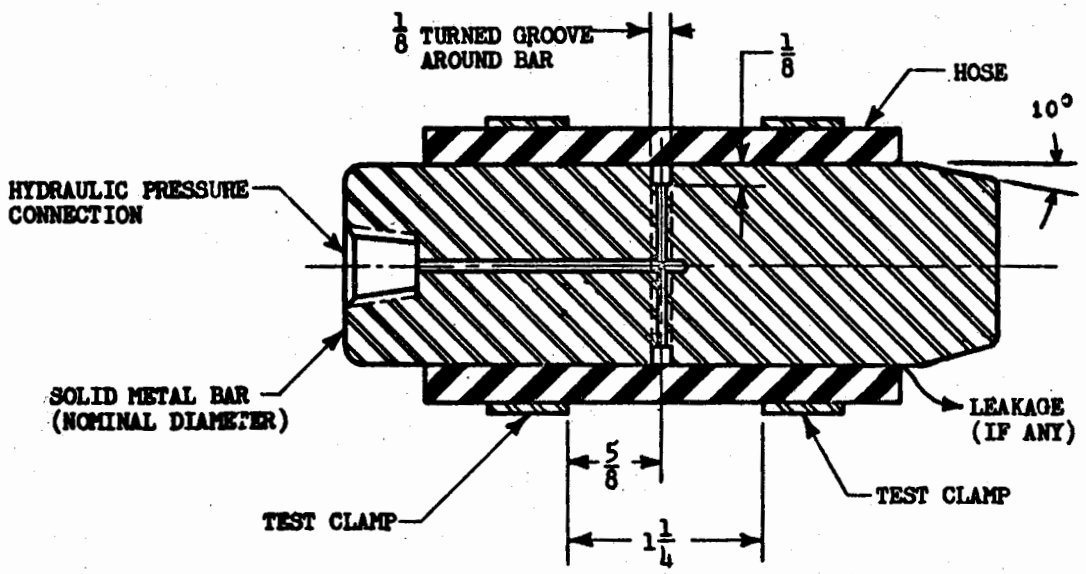


Figure 1. Leakage pressure test set-up

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